



OOP in Java

Slide 1 – Introduction

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What is a Program?

- A program is a set of instructions that tells the computer what to do.
- Computers need instructions written in a language they understand.
- Example: To make the computer display "Hello, World!" on the screen, you write specific instructions.



What is a Java File?

- File as a Container: Imagine your Java file as a box where you store your instructions.
- To keep things organized, you must label the box.
- In Java, this label is called the "class name."

What is the main Part?

- **Entry Point:** The main part of your program acts as the start button.
- When you run the program, the computer looks for this start button to know where to begin.
- Without the start button, the computer doesn't know where to start executing the instructions.
- Similar to turning on a TV: pressing the start button powers it on, and it follows a sequence of instructions to display your favorite channel.

Hello World



HelloWorld.java ×

```
1  ▶ public class HelloWorld {  
2  ▶     public static void main(String[] args) {  
3      System.out.println("Hello, World!");  
4      }  
5  }
```

Hello World

- `public class HelloWorld {`
 - This is like the name tag of the program. We're calling our program HelloWorld. When you create a Java file, the name of the file needs to match this name. So, the file will be saved as HelloWorld.java.
- `public static void main(String[] args) {`
 - This is the start button. Every time we run the program, the computer begins here. It looks for this line to know where to start.
- `System.out.println("Hello, World!");`
 - This is the actual instruction telling the computer to show the words Hello, World! on the screen.
 - It's like telling someone to say "Hello" out loud.
- `}`
 - These curly braces are like bookends. They start and stop the instructions inside your program.

```
% javac HelloWorld.java  
% java HelloWorld
```

Running the Program

- Once the code is written:
 - **Save the file** as HelloWorld.java.
 - **Compile the program**
 - **Run the program:** When you run it, the computer finds the "start button" (main) and prints "Hello, World!" as instructed.

The Movie Script Analogy

- Imagine you're writing a movie script. The script has a title (which is the class name, like "HelloWorld"). The main part is like the beginning of the movie, where the director yells "Action!" The instructions (like `System.out.println("Hello, World!");`) are like the actors speaking their lines.
- When you run the movie (your program), the actors (the computer) follow your instructions, starting with the main part.

- **Program:** A set of instructions for the computer.
- **File Name:** The name of the program, which must match the label (class name).
- **main:** The starting point or "action button" that tells the computer where to begin.
- **System.out.println("Hello, World!");**: The instruction that makes the computer say something on the screen.

Summary

Breaking Down

```
System.out.println("Hello, World!");
```

-
- Imagine you're speaking through a microphone connected to a speaker. In this case, the microphone is the `System.out`, and it sends your words to the speaker, which is the computer screen.
 - Breaking It Down:
 - `System`: Think of it as the whole computer system.
 - `out`: This is like a path or connection that allows the computer to send information to the outside world (the screen).
 - `println`: This stands for "print line," and it means "write the text on a new line."

What is Inside the Parentheses ("Hello, World!")?

- Inside the parentheses, we put the exact words we want the computer to show. It could be any text, like "Welcome to Java!" or "I love programming!".
- The text must be inside **quotation marks** (""). This tells the computer that you are writing text to display, not a command.

Exercise

- `System.out.println("My name is [Your Name].");`
- `System.out.println("I love [Your Hobby].");`
- `println` vs `print`
 - With `println`: The text will appear on one line and the next message will start on a new line.
 - With `print`: The text will appear on the same line as the previous one, without a new line after.